

Software Requirements Specification (SRS)

Skin Cancer Detection App Using Machine Learning

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) specifies both functional and non-functional requirements for the Skin Cancer Detection App Using ML. The project's goal is to develop a mobile application that analyzes skin lesion images using ML and provides users with an automated classification result. In addition to offering a user-friendly interface for submitting skin lesion images for analysis, the application will be designed to assist with initial assessment and raise awareness.

This document describes the intended functionality of the proposed application and outlines the specifications that will guide the project's subsequent design, implementation, integration, and testing stages. It outlines the necessary machine-learning capabilities, user interactions, external interfaces, performance standards, operational limitations, and other considerations throughout development.

The SRS aims to establish a common understanding of the project requirements among the team members, the faculty advisor, client, and other stakeholders. The requirements defined in this document will serve as a foundation for evaluating whether the final application satisfies the specified requirements and goals.

This application will be designed as an assistive and informative tool and is not intended to provide users with a medical diagnosis or to replace evaluation by a qualified healthcare professional. For this reason, the predictions from the ML model will be offered to users for informational and awareness purposes only.

1.2 Scope

Skin Cancer Detection App Using ML will be a mobile application designed to help users track their overall skin health by analyzing images of their skin lesions using a machine-learning classification model. The application will prompt users to create an account and log in before accessing its features. Image analysis will be performed via a server and API, requiring an Internet connection to obtain predictions from the ML model.

The application will:

- Allow users to capture a new skin lesion image or upload an already existing image from a device.
- Classify the submitted skin lesion image as benign or malignant and display the model's confidence score.
- Assign timestamps to submitted images and maintain a chronological history of previous analyses.
- Allow authenticated user to access or delete their stored images and analyzed data.
- Securely transmit and store user images, and restrict access to user data to authorized users.
- Display a disclaimer stating that the prediction of the ML model is not a medical diagnosis.
- Support one mobile platform: Android.

The application will not:

- Provide a formal medical diagnosis.
- Replace the evaluation of a qualified healthcare professional.
- Automatically identify or match the same skin lesion across multiple images.
- Provide medical treatment or make clinical decisions regarding the user.

1.3 Definitions, Acronyms, and Abbreviations

- **AI:** Artificial intelligence. Computer systems designed to carry out tasks that usually require human intelligence.
- **API:** Application Programming Interface. A mechanism that allows a mobile application to communicate with a server for image analysis or other system functions.
- **CNN:** Convolutional Neural Network. A deep learning model commonly used for image classification.
- **ML:** Machine learning. A branch of AI that allows computer systems to learn patterns from data and make predictions.
- **UI:** User interface. The visual components that users interact with in the mobile application.
- **Benign Lesion:** A non-cancerous skin lesion.
- **Malignant Lesion:** A cancerous skin lesion.
- **Confidence Score:** A numerical value representing the confidence level of the ML model regarding the classification result.
- **Inference:** The process of using an ML model to generate a prediction for a new input image.
- **Authentication:** The process of verifying a user's identity before permitting access to their account and stored information.

1.4 References

- IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*, Institute of Electrical and Electronics Engineers (IEEE).
- Skin Cancer Detection App Using ML - Project Plan, Fall 2026.
- Client/Faculty Advisor Requirements and Feedback, Dr. Zahra Nematzadeh, Department of Electrical Engineering and Computer Science, Florida Institute of Technology, Fall 2026.

1.5 Document Overview

The rest of this document is separated into three main sections:

Section 2, Overall Description, provides a general overview of the application, including its product perspective, user features, constraints, assumptions and dependencies, and apportioning of requirements.

Section 3, Specific Requirements, defines the application's functional and non-functional requirements and provides main use case scenarios.

Section 4, Supporting Information, provides additional information regarding the requirements document, including appendices and references to supporting documents.

2. Overall Description

2.1 Product Perspective

Our project, The Skin Cancer Detection App Using ML will be a mobile application that is designed to help users with monitoring their health by allowing users to take a new picture or upload a new picture from their device. Then, these objects will be prepared through preprocessing steps such as resizing and normalization for ML models to analyze. This preprocessing step will help make the image more consistent for the model to learn.

The ML Model will analyze the processed image and share its prediction about skin being benign or malignant. Since these predictions are not %100 correct, the model will also display its confidence level about the prediction. The project will evaluate different ML approaches to determine which model performs better according to the results.

The application will also allow users to save previous results and view the process of their skin lesions history over time. This will allow users to see differences and predictions from different dates instead of using only one analysis to decide. Also, whether it is fully confident or not, these results are only for early detection; they do not replace a medical professional or medical diagnosis. Our app will show appropriate disclaimers and it will tell users to seek professional help if needed.

2.2 Product Functions

The Skin Cancer Detection App will provide several main functions that allow users to upload or take pictures, receive a ML prediction and track previous results over time using a user profile. Main functions:

- **User Account and Login:** This part is important for the saved images and previous analysis results. Users will create an account and log in before accessing the main features of the application.
- **Image Capture and Upload:** Users will be able to upload existing images from their device or take new pictures of skin lesions on the app using a phone camera. This gives users different options to provide an image for analysis.
- **Image Preprocessing:** The application needs this step before learning for the prediction. Before an image is analyzed, the system will prepare it for the ML model. This may include resizing, cleaning and normalizing the image so ML models can perform better and more consistently.
- **Skin Lesion Prediction:** After preprocessing, a ML model will analyze the processed image and classify the skin lesion as benign and malignant. This prediction will be the main goal for this application.
- **Confidence Levels:** Along with the prediction, the application will show the model's confident level on the prediction. This helps users to understand how confident our model is and it gives users more information about the prediction instead of only showing if it is a benign or malignant result.
- **History and Progress Tracking:** Users will be able to save previous skin lesion results and pictures from different days. Also, users can see these past results as well. This will allow users to monitor their previous analyses and see how their results have changed over time.
- **Medical Disclaimer and Next Step:** This application will clearly state that its prediction is not a medical diagnosis. It will provide basic next-step information and encourage users to seek professional medical advice when necessary.

2.3 User Characteristics

The Skin Cancer Detection App is built for general users who want to monitor their skin health and check skin lesions to see if it is malignant or benign. Users do not need any experience in machine learning or the medical field to use the application. The system will be designed to be simple and easy to understand for people with different levels of technical knowledge.

Since users are required to create a profile and upload or take pictures, users are expected to have basic smartphone skills such as taking pictures, uploading pictures, and navigating a mobile application. The prediction and confidence level will be displayed in a simple way so it can be more understandable for users.

2.4 Constraints

The Skin Cancer Detection App has several limitations regarding the platform the app is going to be designed for, quality, and wrong results. These limitations must be considered during development; for example, this application is designed for the Android environment, and it will need to work properly within the supported Android environment

2.5 Assumptions and Dependencies

There are some assumptions for The Skin Cancer Detection App to work properly. Users are expected to have an Android device with a working camera and users should be able to use the app properly. The image should have reasonable lighting and quality so a machine learning model can analyze it correctly.

This project also depends on some datasets like ISIC and HAM10000 for training the model. The development of the system depends on tools and frameworks such as Python, OpenCV, TensorFlow/Keras and Flutter.

The system also assumes that the selected machine learning model can be successfully trained and integrated with the mobile application to provide predictions.

2.6 Apportioning of Requirements

Some features may be added in the future but they are not required for the initial version. Future improvements are:

- **IOS Support:** Future versions may expand the application beyond Android to support IOS devices.
- **Automatic Lesion Matching:** The application could automatically recognize and match the same skin lesion across images taken on different dates.

3. Specific Requirements

3.1 Functional Requirements (FRs)

The functional requirements describe tasks that the system should accomplish to attain the goal of this project. The functional requirements are:

FR-01: Input of an Image

- Users may upload an image via the device's camera or gallery.

FR-02: Image Preprocessing

- Process the image to match the format and requirements of training ISIC datasets.

FR-03: Machine Learning Model Training

- Train CNN models using skin-lesion datasets.

FR-04: Machine Learning Model Evaluation

- Evaluate the trained models using metrics such as accuracy, precision, and recall.

FR-05: Skin Lesion Prediction

- Provide predictions of skin lesions (Benign or Malignant).

FR-06: Prediction Confidence Visualization

- Provide a percentage of how certain the model is regarding prediction accuracy.

FR-07: Comparison with Training Data

- Compares images with thousands of images from the trained datasets.

FR-08: Safety Information and Disclaimers

- Informs users that results are not a medical diagnosis to ensure safety.

FR-09: History and Lesion Tracking

- Tracking the progress of the lesion through multiple image sessions.

FR-10: Data Management

- Store and organize the user's account information and images, and implement security measures.

FR-11: Application Interface

- Provide a simple application interface for easy navigation.

3.2 Non-Functional Requirements (NFRs)

Non-functional requirements describe how well the system performs. These are the non-functional requirements:

NFR-01: Performance

- Preprocess the image and provide the prediction in a short amount of time

NFR-02: Scalability

- Supports additional users, images and different features

NFR-03: Accuracy

- Model accurately identifies unseen images

NFR-04: Reliability

- Consistently produce results without crashing or handling errors

NFR-05: Usability

- Simple interface and intuitive for users with no technical experience

NFR-06: Security

- Protect data from unauthorized access

NFR-07: Maintainability

- Reliable code and structures to ensure easy modification of code if needed

3.3 Use Case Scenarios

UC-01: User creates an account

- Users have the option to create an account or sign in to keep track of image sessions. Users may skip this step and enter the app as a guest (history will not be saved).

UC-02: Uploads/Captures Image

- If a user uploads an image, the app will preprocess the image for a couple of seconds.
- If the user decides to capture the image via the camera device, the app interface will ask for camera permission. After permission is granted, the user can take a photo and upload it. After the image is uploaded, it will be preprocessed.

UC-03: Analyze Skin Lesion

- The machine learning model analyzes preprocessed images to generate the prediction.

UC-04: Display Prediction

- Prediction with confidence level is displayed.
- If the uploaded image has poor quality due to lighting or blurriness, or other factors that can affect the quality, the app will ask the user to rescan the image.

- If the prediction is benign with a low confidence level (ex: <50%), the app will provide next steps.
- If the prediction is malignant with high confidence (ex: >80%), the app displays a warning message to immediately seek medical attention and provides next steps to guide the user to safety.

UC-05: Safety Information

- The safety information is delivered after the prediction is displayed. Depending on the prediction, the app may provide important links for the user or next steps for the user to continue monitoring lesions.

4. Supporting Information

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4.1.2 Index

- Account Management & Authentication: 1.2, 1.3, 2.2, 3.1 (FR-10), 3.3 (UC-01)

- Accuracy & Performance Metrics: 1.3, 2.1, 3.1 (FR-04), 3.2 (NFR-01, NFR-03)
- Android Platform Constraints: 1.2, 2.4
- API (Application Programming Interface): 1.2, 1.3
- Artificial Intelligence (AI) & Machine Learning (ML): 1.1, 1.2, 1.3, 2.1, 2.2, 3.1 (FR-03, FR-05)
- Benign Lesion: 1.2, 1.3, 2.1, 2.2, 3.1 (FR-05), 3.3 (UC-04)
- Confidence Score/Visualization: 1.2, 1.3, 2.1, 2.2, 3.1 (FR-06), 3.3 (UC-04)
- Convolutional Neural Network (CNN): 1.3, 3.1 (FR-03)
- Data Privacy & Security: 1.2, 3.1 (FR-10), 3.2 (NFR-06)
- Guest User Access: 3.3 (UC-01)
- Image Preprocessing & Normalization: 1.3, 2.1, 2.2, 3.1 (FR-02), 3.3 (UC-02)
- Image Quality & Rescan Prompt: 3.3 (UC-04)
- ISIC Dataset: 3.1 (FR-02, FR-07)
- Lesion History & Progress Tracking: 1.2, 2.1, 2.2, 3.1 (FR-09)
- Malignant Lesion: 1.2, 1.3, 2.1, 2.2, 3.1 (FR-05), 3.3 (UC-04)
- Medical Disclaimer & Safety Information: 1.1, 1.2, 2.1, 2.2, 3.1 (FR-08), 3.3 (UC-05)
- Reliability & System Availability: 3.2 (NFR-04)
- Scalability: 3.2 (NFR-02)
- User Interface (UI) & Usability: 1.3, 2.3, 3.1 (FR-11), 3.2 (NFR-05)

4.2 Appendices

4.2.1 Appendix A: Client & Stakeholder Engagement Plan

- Initial Project Orientation & Scope
 - Participants: Dr. Zahra Nematzadeh (Client/Faculty Advisor), Samuel Borges, Elif Dila Bayrak, Neva Ildem, Sophia Zubieta.
 - Baseline Requirements: Initial Scope set for Android, focusing on binary classification (Benign vs. Malignant) with confidence score visualization, including a user account and safety disclaimers
- Pending Client Requirements Review:
 - Objective: Formal review session with Dr. Nematzadeh to evaluate SRS draft prior to further development

4.2.2 Appendix B: Preliminary Technical Validation and UI Wireframes

- Backend/Model Pipeline:
 - Preliminary testing has been initiated on the core machine learning pipeline. Initial testing focuses on evaluating using a standard

Convolutional Neural Network (CNN) compared to a Transfer Learning model to compare accuracy prior to final model selection.

- Frontend/User Interface Concepts:
 - Screen 1: Conceptual layout for account login or registration with medical disclaimer acknowledgement
 - Screen 2: Interface design to select or capture photo
 - Screen 3: Loading bar with quality validation for photo; is able to prompt rescan if necessary
 - Screen 4: Display classification outcome as benign or malignant, displays model confidence score, and potential next steps or medical preventions and disclaimers
 - Screen 5: Asks if the user wants to save as part of the history log, and shows tracking of other past uploads and timestamps for users with an account.

4.3 References to Supporting Documents

1. IEEE Computer Society, “*IEEE Recommended Practice for Software Requirements Specifications*,” IEEE Std 830-1998, IEEE Standards Board, 1998.
2. International Skin Imaging Collaboration (ISIC), “*ISIC Archive for Skin Lesion Analysis and Image Standardization*,” Public Dataset Resource for Machine Learning Model Training and Benchmark Evaluation. [Online]. Available: <https://www.kaggle.com/datasets/nodoubttome/skin-cancer9-classesisic/data>
3. Nematzadeh, Dr. Zahra, “*Client/Faculty Advisor Requirements and Feedback*,” Department of Electrical Engineering and Computer Science, Florida Institute of Technology, Fall 2026.
4. Senior Design Project Team (Samuel Borges, Elif Dila Bayrak, Neva Ildem, Sophia Zubieta), “*Skin Cancer Detection App Using ML - Project Plan*,” Department of Electrical Engineering and Computer Science, Fall 2026.
5. Senior Design Team Web Repository, “*Softcopy Deliverables, User Interface Wireframes, and System Documentation*,” Hosted on Official Project Website, Fall 2026.[Online]. Available: <https://znildem.github.io/index.html>